Methods of Molecular Simulation and the Applications for Methane Under Extreme Conditions

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Abstract

Matter in extreme conditions is difficult and expensive to test physically. Due to the complicated nature of all but the simplest of molecules, the equations of state are not known. Computer simulations provide a method to determine the equations of state under extreme conditions. This paper will focus on two widely used simulation methods for methane in the solid and liquid states, the theory behind the simulations and also briefly touch on the applications of the results.

1 Introduction

Dense states of methane are found in celestial bodies where conditions subject matter to high pressures and low temperatures. These conditions are favourable for the formation of solid and liquid methane. Saturn's largest moon, Titan, is well known for methane lakes and its methane-based hydrological cycle [1]. Near-infrared spectroscopic observations of the icy surface of celestial bodies also suggest the presence of solid methane on Pluto [2]. Methane is essential for the formation of other organic molecules, and understanding its behaviour outside of Earth may provide insights into fields such as astrobiology. Solid methane also has applications in neutron moderation, which is a process of slowing down fast neutrons used to propagate nuclear chain reactions [3].

However, the behaviour of solid and liquid states of methane is not well understood under extreme conditions of high pressures and low temperatures. Currently, there are no analytical solutions to accurately predict the behaviour of methane. Hence, simulations are essential to test current models against experimental observations. Furthermore, simulations allow properties of methane to be studied under experimentally infeasible conditions. This paper aims to explore how simulations are conducted and how the structure of methane adds a layer of complexity to accurate simulations of dense states of methane. The link between macroscopic measurements can be made from the microscopic properties in simulations. Modelling the potential energy field between methane molecules is an area of research and is crucial to extracting meaningful macroscopic information from simulations. We end the paper with a brief discussion of the applications of dense states of methane in space and neutron moderation research.

2 Macroscopic and Microscopic Thermodynamics of Simulations.

From an experimental point of view, macroscopic variables such as temperature (T), pressure (P), and volume (V) are measured to determine conditions for phase changes. However, experiments under extreme conditions may be infeasible, and the equations of state for complex molecules are not well-defined. Simulations require a microscopic understanding of the system to enable these macroscopic properties to be determined indirectly.

2.1 Macroscopic Considerations in Simulations.

Macroscopic constraints need to be imposed on the system before running simulations. In an NVT ensemble, the number of molecules, temperature and volume of the system are controlled, and the Helmholtz free energy (F) is the natural thermodynamic potential that characterizes the system. In an NPT ensemble, the Gibbs free energy is used when the pressure is controlled instead of the temperature [4].

The second law of thermodynamics requires the systems at equilibrium to maximize entropy and minimize free energies. The state with the lower free energy is significantly more probable [5]. Coexistence occurs when the free energies of the two states are equal [4].

By imposing specific thermodynamic constraints, the simulation for each phase is allowed to evolve until equilibrium when the free energy is minimized. By adjusting a single thermodynamic variable (x), the function of the free energy can be plotted, as in Fig. 1. From the plots, the equilibrium conditions can be extracted and compared to real-life mea-

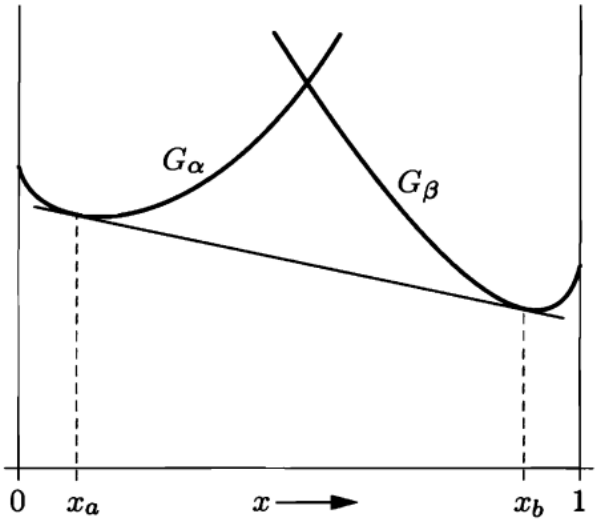


Figure 1: Plot of the free energy of two different states against a free thermodynamic variable x. The gradient of tangent line construction between two phases of a material α and β gives the equilibrium conditions for another variable of interest [6].

surements or used as a reference for future experiments, as in Fig. 7.

An example used in this report is determining the melting line parameters of methane. This can be done using tangent line construction, as seen in Fig. 1. By keeping the temperature and number of molecules constant but simulating a set of volumes until equilibrium, a set of Helmholtz free energy curves can be plotted for solid and liquid methane. The gradient of the tangent line construction would give the equilibrium pressure of the system.

2.2 Microscopic Considerations in Simulations

In statistical mechanics, a microstate is defined as the position and momenta of the individual particles of a system, $\{x_i;p_i\}$ and a macrostate is defined by the thermodynamic macroscopic variables of the system, $\{N,\ V,\ T,$ etc.$\}$, [5]. The energy E_i of the system is distributed by a Boltzmann distribution as seen in Eq. 1, which is expressed with the partition function Z and the Boltzmann factor k_b.

$$P_i = \frac{e^{-\frac{E_i}{k_b T}}}{Z} = \frac{e^{-\frac{E_i}{k_b T}}}{\sum_i e^{-\frac{E_i}{k_b B T}}} \tag{1}$$

Simulations iterate through possible microstates of the system in an attempt to minimize the free energy. The exact Helmholtz free energy and Gibbs free energy can be evaluated with Eq. 2 and 3 using the partition function. However, this requires simulating the system for an infinite amount of time, which is not possible. Hence, alternative means are used to evaluate the free energies, such as thermodynamic integration and free energy perturbation, which will not be discussed in this report [7].

$$F = -k_b T \ln Z \tag{2}$$

$$G = -k_b T \ln Z + PV \tag{3}$$

Later in the report, it will be seen that the total energy is a quantity of interest when evolving the system. It can be trivially evaluated as the sum of the potential and kinetic energies. Kinetic energy can be easily computed from the molecule's velocity. However, greater care is needed to evaluate potential energy because the potential field is not well defined.

3 Modelling Potentials for Solid and Liquid Methane.

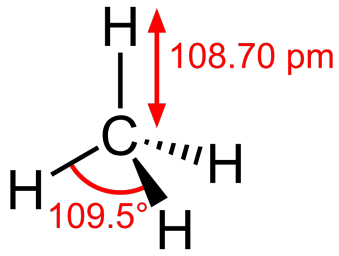


Figure 2: Molecular structure of a methane molecule with a tetrahedral structure with a central carbon atom and four hydrogen atoms [8].

Determining the scalar potential field of methane is an area of active research. In simulations, the scalar potential plays a key role in the evolution of the system. The structure of methane is shown in Fig. 2, the molecule has no net charge, but the multipole expansion has a non-trivial octupole term. The potential energy of a collection of methane molecules is commonly approximated with the short-range Lennard-Jones Potential and the long-range Coulomb contributions from the octupole moment. Dispersion forces are present due to random quantum excitations resulting in instantaneous dipoles that influence the potential field [9]. Determining the potential field is not as trivial as it seems. In this report, only the contribution from the Lennard-Jones Potential and the octupole contributions are discussed.

3.1 The Lennard Jones Potential

The Lennard-Jones (LJ) Potential given in Eq. 4 describes the interaction between a pair of molecules separated by a distance r as seen in Fig. 3. The depth of the potential well (ϵ) and the distance at which the potential is zero (σ) characterises the LJ potential. Some theoretical studies in the literature trivially consider the distances between the carbon centres of methane molecules to compute the LJ potential [10]. Other simulations utilize a 5-atom model to simulate methane molecules [11, 12]. The sum is over pairs of methane molecules, α and β, are considered in Equation 4 with i and j as the atoms within each molecule [11].

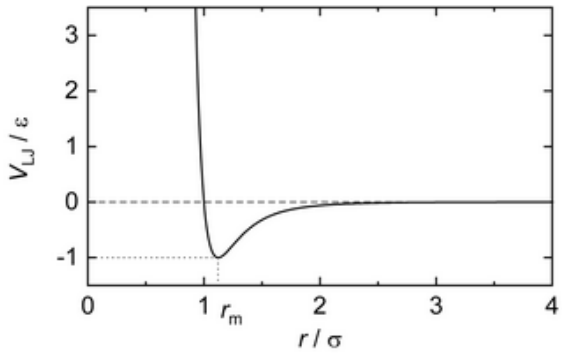


Figure 3: Graph of the Lennard Jones Potential. The positive repulsive term dominates at short distances while the negative attractive term dominates at larger distances [13].

$$U_{\mathrm{LJ}}(r) = \sum_{\alpha,\beta} \sum_{i\in\alpha, j\in\beta} \left[4\epsilon_{ab} \left(\frac{\sigma_{ab}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ab}}{r_{ij}} \right)^{6} \right] \qquad (4)$$

The LJ potential is an efficient and easy model to implement but it fails to provide good estimates for structural properties of dense fluid methane [11]. To accurately model dense states of methane, a more accurate description of the potential field is required.

3.2 Electrostatic Octupole Interactions.

The tetrahedral nature of methane causes the dipole and quadrupole moments to vanish in its multipole expansion, leaving a nontrivial contribution from the octupole moment. A detailed computation of the octupole and hexa-decapole moments of methane is provided by Amos [14]. Octupole moments are dependent on the orientation of methane molecules in space, as shown in Fig.4. In fluid simulations, octupole moments are evaluated by directly summing up the Coulomb potentials and adding this contribution to the LJ-potential. This is known as the Optimized Potentials for Liquid Simulations, [11, 12], as seen in Eq. 5.

$$U(r) = \sum_{\alpha,\beta} \sum_{i\in\alpha, j\in\beta} \left[U_{LJ}(r) + \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} \right] \qquad (5)$$

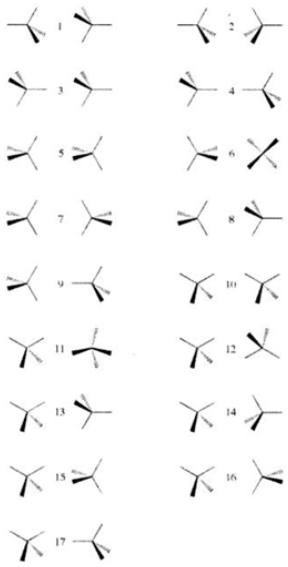


Figure 4: Orientations of methane in spherical space calculated in the paper 'Ab initio intermolecular potential energy surface and second pressure virial coefficients of methane' [15].

3.3 Lattice structure

In the solid phase, methane molecules adopt a well-defined crystalline arrangement. Experimental studies revealed that solid methane exhibits distinct phases depending on temperature and pressure [16]. In the high-temperature Phase I, methane molecules are arranged in a face-centred cubic (FCC) lattice with rotational disorder. As the temperature decreases, methane transitions to FCC Phase II at 20.4K [17], with orientationally ordered molecules as seen in Fig.5. The total lattice energy can be approximated by summing the pairwise interactions over all the molecular pairs. The random orientation of methane molecules in Phase I allows the first-order terms of the octupole interaction to vanish, while in FCC Phase II the first-order terms of the octupole contributions remains non-trivial [10].

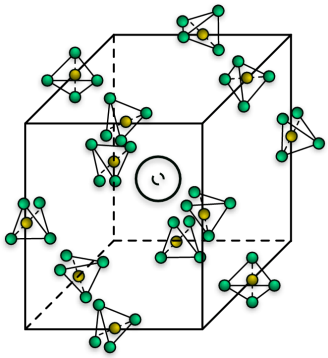


Figure 5: Structure of solid methane in phase II, figure from [16].

4 Methods of Simulation.

4.1 Why Simulations?

Due to the uncertain nature of the potential field present in a sample of methane, the exact equations of state are unknown. This has led to various proposals for potential functions which need to be tested in some way. Real experiments involving methane in extreme conditions are difficult and expensive to produce in the real world so alternatives need to be sought out. Computer simulations take advantage of already well known physics (Newtons laws, Boltzmann distribution) to attempt to model the behaviour of the system numerically. Various models exist, all of which make their own approximations to ensure the simulation concludes in a reasonable amount of time. These approximations have to be carefully selected to ensure accurate results. For example, computationally expensive functions can be series expanded, the total number of particles can be restricted, or the tolerances can be relaxed. Two families of techniques will be considered in this report - Monte Carlo and Molecular Dynamics.

4.2 Monte Carlo Methods

Monte Carlo methods are a wide umbrella of techniques which use random sampling to solve problems, [18]. Specifically in this case the Markov Chain Monte Carlo method is an algorithm used to explore the state space by randomly rearranging particles (unphysically) and checking if the new configuration is favourable [19]. This process does not need to take a physical path because the entire state volume explored is independent of the algorithm used to explore it - this is called Liouville's theorem [20]. A visualization of this algorithm, and the molecular dynamics algorithm, can be seen in Fig. 6.

For a single microstate, defined in Section 2.2, the algorithm updates as follows: The positions and momenta of the particles in the microstate are updated randomly to a new microstate in the state space. In the new microstate, s_{new}, the total energy, U_{new}, is calculated by summing each particle's potential and kinetic contributions. Next, the simulation has to decide whether to accept or reject this new microstate. This is determined by a probability calculated using the Boltzmann distribution, Eq. 1. The original proposal by Metropolis et al., [19], was to accept or reject the new microstate with a probability of:

$$P_{accept}(s_{new}) = \begin{cases} \frac{N_{\text{new}}}{N_{\text{old}}} & N_{\text{new}} < N_{\text{old}} \\ 1 & N_{\text{new}} \geq N_{\text{old}} \end{cases} \tag{6}$$

where N represents the number of possible microstates with a given energy U. This condition has survived the test of time and is still the dominant criteria used in simulations today, [7].

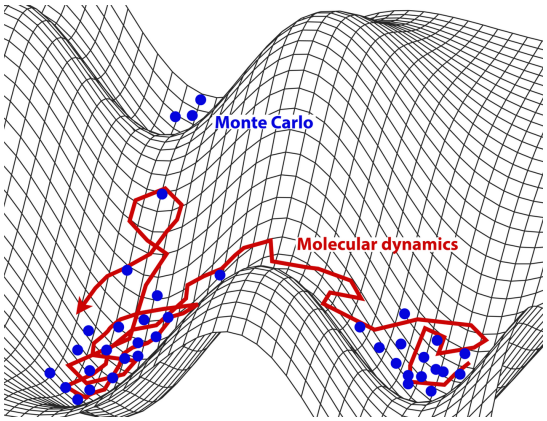


Figure 6: A visualisation of the Monte Carlo and Molecular Dynamics algorithms exploring a state space. The blue dots represent the position evolution of a single particle in the Monte Carlo algorithm and the red line shows the path of a single particle in a molecular dynamics evolution. Note that the blue dot does not take a physical path between positions. Figure from [21].

4.3 Molecular Dynamics Methods

Another family of algorithms for exploring the state space are the Molecular Dynamics methods. This family of algorithms rely on Newton's equation of motion and attempt to model real dynamics. Because the full equations of motion in the potential field are very computationally expensive they are approximated. For example, in the Verlet algorithm the position and velocity are Taylor expanded around a small time displacement Δt:

$$r\left(t \pm \Delta t\right) = r(t) \pm v(t)\Delta t \pm \frac{F(t)}{2m}\Delta t^2 + \frac{\Delta t^3}{3!}\frac{d^3 r}{dt^3} + O(\Delta t^4)$$

and after some algebra expressions for $r(t)$ and $v(t)$ are obtained, [7]:

$$r\left(t + \Delta t\right) \approx 2r(t) - r(t - \Delta t) + \frac{F(t)}{m}\Delta t^2 \tag{7}$$

$$v(t) = \frac{r(t + \Delta t) - r(t - \Delta t)}{2\Delta t} + O(\Delta t^2) \tag{8}$$

The system evolves through time keeping track of the microstates until the algorithm arrives at a minimum value for the chosen free energy in the state space.

5 Applications.

5.1 Determining Phase Change Parameters for Methane.

Molecular simulations verify theoretical predictions and experimental results as seen in Fig 7. Theoretical equations of state proposed by Yakub et al. for solid and liquid methane were used to determine the melting line parameters for methane

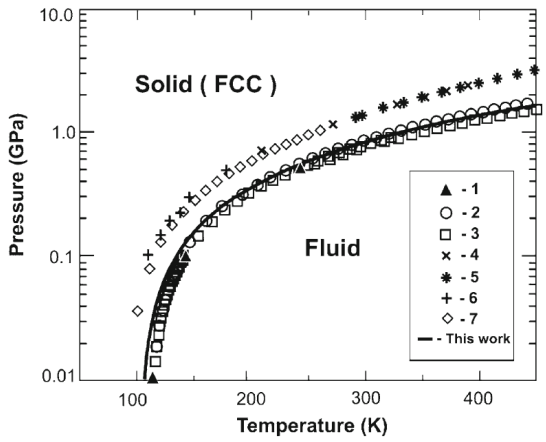


Figure 7: Comparison of theoretical predictions (solid line), Monte Carlo simulation (1 - 3) and experimental results (4 - 7) [10].

[10]. The results show that the predicted melting line parameters agree well with simulation data. However, there is a systemic deviation from the actual experimental data. This highlights that there is an incomplete understanding of the behaviour of methane and the authors propose that it is due to the inadequate description of the short-range repulsion of molecules. Hence this is an example of how simulations can be used to verify our understanding of molecular systems and approximate melting line parameters without experimentation.

5.2 Applications in Space Research

The research on methane under extreme conditions helps the development of space science. Methane, as one of the main components of the atmosphere of Saturn's moon Titan, plays an important role in maintaining the abundance of nitrogen in Titan's atmosphere [22]. Although the presence of nitrogen (N_2) and methane (CH_4) limits the visible light reaching the surface of Titan, lakes of methane and ethane (C_2H_6) are detected by the Visual and Infrared Mapping Spectrometer (VIMS) on the probe Cassini. This discovery, together with the study of phase changes of methane provides a better understanding of Titan's atmosphere [1]. Furthermore, later detections and analysis indicate the existence of exposed ice-rich bedrock on the surface of Titan, which implies the activities of liquid methane [23].

Methane was first identified with reflectance spectrum at 1.7 and 2.3 μm on Neptune's moon Triton and Pluto, but its physical state was unclear [24]. This was later believed to be solid methane by comparing it with laboratory-produced spectrum and further confirmed by better detection methods. In short, theoretical analyses together with computer simulations and experiment results of condensed methane on its melting line have enhanced celestial bodies research.

5.3 Applications in Neutron Production.

With simulation data, optimum conditions to maintain solid methane in industrial systems can be determined and studied. Solid methane is utilised in neutron moderation, the process used in nuclear reactors, in which fast-moving neutrons are slowed due to elastic scattering with a material. Neutrons are slowed down due to the increased probability of interaction with the nuclei instead of phasing through the target. Studies show that solid methane is an effective moderator due to the large proportion of hydrogen atoms [25]. The weight of a hydrogen atom is similar to the weight of a neutron, resulting in a larger proportion of the neutron's kinetic energy being transferred to the hydrogen atom compared to heavier atoms. The efficiency of methane as a moderator is 3 times higher than that of liquid hydrogen [26].

Hydrogen has a scattering cross-section higher than any other nuclide, making methane's high density of hydrogen an effective moderator [27]. The scattering cross-section influences the likelihood that a neutron will scatter upon interaction with the chosen nucleus. A higher scattering cross section means a larger effective area for the neutron to travel through to generate an interaction, thus slowing the neutron.

However, maintaining solid methane presents challenges as it requires very low temperatures to remain in a solid state at around -183 °C [28]. With molecular simulations, optimum parameters to maintain specific phases of solid methane can be determined with minimal experimental work.

6 Conclusion

In conclusion, simulations provide an effective means to study systems under conditions that are not feasible for experimentation. This paper shows how Monte Carlo and Molecular Dynamic simulations can be used to determine the properties and behaviours of methane under extreme conditions. With simulations, methane hydrological cycles on Titan can be studied. Simulation data also provides an effective method to calculate the conditions required to maintain solid methane in industrial systems and nuclear technologies. Due to the molecular structure of methane, modelling the potential is complicated and needs to be carefully accounted for during simulations. Hence, simulations are only as good as our current understanding of the behaviour of the scalar potential field of methane. With refinements, the agreement between simulation and experimental data can be improved. Hence, simulations are still an important means of deriving meaningful information about the behaviour methane under extreme conditions.

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